

DSPO: STABLE AND EFFICIENT POLICY OPTIMIZATION FOR AGENTIC SEARCH AND REASONING

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ABSTRACT

LLMs require active access to external knowledge to solve complex question-answering problems and related tasks whose evidence lies beyond parametric memory. Existing methods either rely on prompting to elicit latent tool-use behavior or, when trained with reinforcement learning in interactive search environments, encounter performance ceilings and training collapse. These limitations leave the agentic search capability of LLMs systematically under-optimized. We introduce **Dynamic-filter Sequence-level Policy Optimization (DSPO)**, a reinforcement-learning algorithm that stabilizes agent training by coupling sequence-level policy updates with dynamic outcome-based sample filtering. Without supervised demonstration data, DSPO learns to interleave multi-turn search with reasoning using environmental feedback alone. Across multiple QA benchmarks, our 7B model improves over comparable prior work by **34.1%**; on complex multihop QA such as HotpotQA, it further surpasses the 14B model from prior work by nearly **9% relative**, while preserving stable training dynamics.

1 INTRODUCTION

Large Language Models (LLMs) (Brown et al., 2020; Touvron et al., 2023; Zhao et al., 2023) now deliver strong results in specialized domains such as math (Shao et al., 2024; Trinh et al., 2024; Yu et al., 2025), coding (Zheng et al., 2023; Yang et al., 2024), and creative writing (Chakrabarty et al., 2024; Marco et al., 2024). Their factual competence, however, remains constrained by the information available during pretraining. A direct way to expand this competence is to equip LLMs with search, converting them into agents that can query external knowledge sources during problem solving (Jin et al., 2025b). This setting instantiates tool-calling (Schick et al., 2023): the model must determine when to invoke a search tool, formulate an appropriate query, and incorporate retrieved evidence into subsequent reasoning. Because these decisions unfold over long interaction trajectories, agentic search is naturally formulated as a reinforcement-learning (RL) problem over multi-step policies.

RL is particularly well suited to this setting because, unlike Supervised Fine-Tuning (SFT), it can improve policies through exploration and environmental feedback without requiring costly static demonstrations (Chu et al., 2025). Value-free methods such as Group Relative Policy Optimization (GRPO) (Shao et al., 2024) are therefore attractive because they avoid a separately trained value model and reduce memory overhead. Yet methods that are effective for more constrained reasoning tasks become fragile in open-ended interactive search (Jin et al., 2025b; Yu et al., 2025; Cui et al., 2025; Liu et al., 2025). Two failure modes dominate this regime. First, as identified by Zheng et al. (2025), GRPO optimizes token-level ratios while receiving sequence-level rewards, creating high-variance gradients over long trajectories. Second, sparse search rewards frequently yield homogeneous rollout groups in which every sampled trajectory is either successful or unsuccessful; in those groups, the normalized advantage collapses and the update contributes little or no learning signal (Yu et al., 2025; Liu et al., 2025).

We address these coupled sources of instability and sample inefficiency with **Dynamic-filter Sequence-level Policy Optimization (DSPO)**. DSPO unifies two recent policy-optimization principles within a single agent-training procedure. It adopts the sequence-level optimization of GSPO (Zheng et al., 2025), thereby aligning the unit of policy update with the unit at which reward is assigned, and it incorporates a dynamic outcome-based filtering mechanism inspired by DAPO (Yu et al., 2025), retaining only rollout groups that contain both successful and unsuccessful trajectories. This design yields informative group-relative advantages $\hat{A}_i$ while avoiding the token-level variance that destabilizes long-horizon agent training. Although developed for multi-turn search and reasoning, the algorithm operationalizes a broader lesson supported by our experiments: stable RL for LLM agents requires both reward-aligned optimization and samples that contain usable contrast. With this design, our model achieves a **34.1% relative improvement** over a

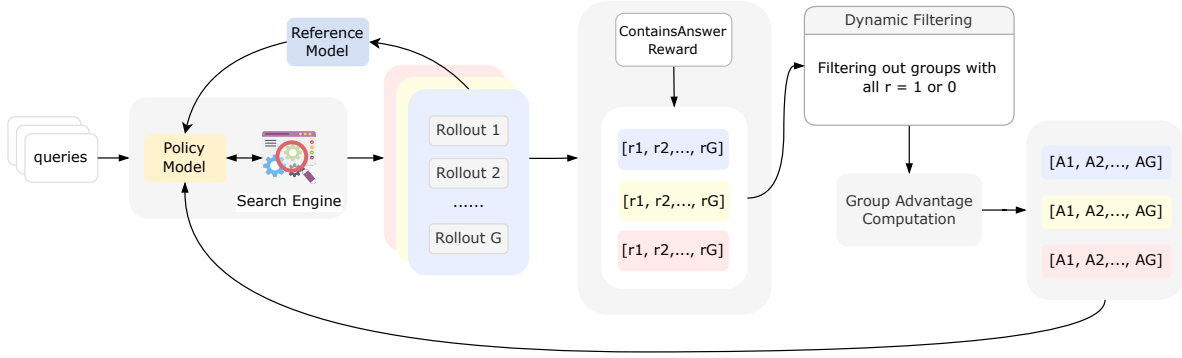


Figure 1: Overview of the DSPO training loop. For each query, the policy samples a group of G search-interaction trajectories and assigns each trajectory a sparse terminal reward. The **dynamic filter** rejects homogeneous-outcome groups and continues sampling until the batch contains informative success–failure contrasts, after which the retained trajectories update the policy through the sequence-level objective.

leading 7B baseline (Jin et al., 2025b) and surpasses its 14B counterpart (Jin et al., 2025a) on the multihop HotpotQA benchmark by **nearly 9% relative** (0.613 vs. 0.563).

This paper makes three concrete contributions.

- We introduce DSPO, an RL algorithm for agentic search models that directly targets the two principal obstacles in sparse-reward interactive search: unstable token-level updates and uninformative homogeneous rollout groups.
- We show that combining **sequence-level optimization** with **dynamic outcome-based filtering** produces an effective training signal for multi-turn search, yielding stable policy updates without supervised demonstration data.
- We establish DSPO’s empirical advantage on seven QA benchmarks: the 7B DSPO model obtains a **34.1% relative improvement** over a comparable 7B baseline, exceeds the corresponding 14B baseline on HotpotQA by **nearly 9% relative** (0.613 vs. 0.563), and exhibits stable reward and validation dynamics under a controlled BM25-retrieval setup.

2 RELATED WORK

2.1 RL FOR LLMs

RL methods for LLMs have progressed from Reinforcement Learning from Human Feedback (RLHF) with PPO and explicit reward models (Ouyang et al., 2022; Christiano et al., 2017; Schulman et al., 2017) toward simpler direct-optimization frameworks such as DPO (Rafailov et al., 2023). Group Relative Policy Optimization (GRPO) (Shao et al., 2024) marks an important value-free direction: it eliminates the value model by estimating advantages from group-level reward statistics. Its token-level objective, however, is known to induce training instability (Liu et al., 2025; Cui et al., 2025), motivating several refinements. GSPO replaces token-level ratios with a sequence-level objective matched to the reward unit (Zheng et al., 2025), whereas DAPO improves sparse-reward learning through dynamic outcome-based sampling (Yu et al., 2025). GMPO similarly uses geometric-mean aggregation to reduce token-level sensitivity to outliers (Zhao et al., 2025). Our experiments show that these ideas remain incomplete when applied separately to autonomous search agents, where collapse and performance bottlenecks persist. DSPO therefore consolidates sequence-level optimization and dynamic filtering into one policy-optimization algorithm designed for the coupled instability and sample-efficiency demands of interactive search.

2.2 LLMs WITH AGENTIC RETRIEVAL

Retrieval-Augmented Generation (RAG) mitigates the static-knowledge limitations of LLMs by integrating external retrievers that supply dynamically updated information (Lewis et al., 2020; Gao et al., 2023). Classical RAG systems retrieve relevant documents, often with dense retrievers, and concatenate them to the model input before generation

(Karpukhin et al., 2020). Such pipelines improve factual access but typically impose a fixed retrieval procedure, limiting autonomy in complex multi-turn scenarios. Recent work has therefore moved toward agentic paradigms in which LLMs plan, search, and reason iteratively. ReAct couples reasoning and acting to support tool interaction in tasks such as web navigation (Yao et al., 2023), and multi-agent systems such as AutoGen organize collaborative workflows (Wu et al., 2024). A further line of work integrates agentic RAG with RL, allowing retrieval behavior to be learned through decision-making. Wu et al. (2025) introduce Agentic Reasoning as a framework for tool-supported LLM reasoning, and RL-based approaches (Jin et al., 2025b; Chen et al., 2025; Song et al., 2025) train LLMs to interleave reasoning and search without supervised demonstrations. This end-to-end paradigm internalizes agent capabilities and reduces the engineering burden of multi-agent pipelines, but it still suffers from instability and performance ceilings in open-ended search. DSPO directly targets these bottlenecks by providing a stable and sample-efficient policy-optimization framework for learning multi-turn search strategies.

3 METHODOLOGY

This section formalizes agentic search as an RL problem, reviews the policy-optimization objectives that motivate our design, and identifies the specific limitations that arise in sparse-reward interactive search. We then present **Dynamic-filter Sequence-level Policy Optimization (DSPO)**, specifying how its sequence-level update and dynamic outcome-based filtering jointly improve training stability and sample efficiency before giving the integrated training algorithm.

3.1 PRELIMINARIES

Policy Gradient Methods for LLMs. RL training for LLMs commonly uses policy-gradient methods such as PPO (Schulman et al., 2017), a standard algorithm for alignment. PPO updates a policy π_θ by maximizing a clipped surrogate objective computed from samples generated by an old policy $\pi_{\theta_{\text{old}}}$. Averaged over tokens, the objective is:

$$J_{\text{PPO}}(\theta) = \mathbb{E}_{x \sim \mathcal{D}, y \sim \pi_{\theta_{\text{old}}}(\cdot|x)} \left[\frac{1}{|y|} \sum_{t=1}^{|y|} \min \left(r_t(\theta) \hat{A}_t, \text{clip}(r_t(\theta), 1 - \epsilon, 1 + \epsilon) \hat{A}_t \right) \right], \quad (1)$$

where $r_t(\theta) = \frac{\pi_\theta(y_t|x, y_{<t})}{\pi_{\theta_{\text{old}}}(y_t|x, y_{<t})}$ denotes the token-level importance ratio. PPO must additionally train a value model to estimate token-level advantages $\hat{A}_t$ through Generalized Advantage Estimation (GAE) (Schulman et al., 2015); this auxiliary model increases memory cost and can itself introduce instability.

GRPO (Shao et al., 2024) removes this value-model dependency by sampling a group of G responses $\{y_i\}_{i=1}^G$ for a prompt x and normalizing each response reward against the group’s reward statistics. Despite this value-free advantage estimation, the optimization objective remains token-level:

$$J_{\text{GRPO}}(\theta) = \mathbb{E}_{x \sim \mathcal{D}, \{y_i\}_{i=1}^G \sim \pi_{\theta_{\text{old}}}(\cdot|x)} \left[\frac{1}{G} \sum_{i=1}^G \frac{1}{|y_i|} \sum_{t=1}^{|y_i|} \min \left(r_{i,t}(\theta) \hat{A}_i, \text{clip}(r_{i,t}(\theta), 1 - \epsilon, 1 + \epsilon) \hat{A}_i \right) - \beta D_{KL}(\pi_\theta || \pi_{\text{ref}}) \right], \quad (2)$$

where $r_{i,t}(\theta) = \frac{\pi_\theta(y_{i,t}|x, y_{i,<t})}{\pi_{\theta_{\text{old}}}(y_{i,t}|x, y_{i,<t})}$, and the advantage for every token $y_{i,t}$ in a response y_i is set to the same sequence-level value:

$$\hat{A}_{i,t} = \hat{A}_i = \frac{R_i - \text{mean}(R)}{\text{std}(R)}, \quad (3)$$

Thus, every token in response y_i is assigned the same advantage $\hat{A}_i$, even though the clipping and importance correction are applied at token resolution. This mismatch between sequence-level reward assignment and token-level optimization is the instability that DSPO resolves.

Agentic Search as a Markov Decision Process. We formulate iterative agentic search and reasoning as a discrete-time, finite-horizon Markov Decision Process (MDP), in which the policy alternates between reasoning, tool invocation, and final-answer generation.

- **State (s_t):** At turn t , the state s_t contains the full interaction history: the initial question q , all previously generated reasoning and search queries, and the retrieved evidence returned by the environment. Because

each retrieval result must be interpreted and incorporated into later decisions, the state expands over the trajectory and creates long-horizon dependencies that are poorly matched to conventional token-level RL updates.

- **Action** (a_t): An action is a complete textual segment generated by the policy π_θ , consisting of free-form reasoning followed by a control decision. The segment terminates either with a `</tool_call>` token, which invokes the search engine and inserts the returned evidence within `</tool_response>`, or with a `</answer>` token, which ends the trajectory. This structured action space requires the agent to learn both *what* content to generate and *when* external search is necessary, producing substantial variation in trajectory length.
- **Policy** (π_θ): The policy π_θ is the underlying LLM, which generates tokens autoregressively conditioned on s_t . Although the model conditions on the complete interaction history, the optimization target is computed only over model-generated reasoning and action tokens; retrieved environment content is masked out (Jin et al., 2025b). Learning this policy therefore requires credit assignment across long sequences in which intermediate reasoning steps receive no direct supervision, reinforcing the need for stable sequence-level optimization.
- **Trajectory** (τ): A trajectory $\tau = (s_1, a_1, \dots, s_T, a_T)$ records the complete reasoning and tool-interaction process for a question, including model-generated actions and search results returned by the environment. Since the reward is assigned only after the final answer, the learning problem is inherently sequence-level: early query choices and reasoning decisions can determine eventual answer correctness.
- **Reward** ($R(\tau)$): We use a sparse terminal reward. A trajectory receives $R = 1$ when the final answer contains the ground-truth text and $R = 0$ otherwise:

$$R(\tau) = \begin{cases} 1 & \text{if } a_{\text{gold}} \subseteq a_{\text{pred}}, \\ 0 & \text{otherwise.} \end{cases} \quad (4)$$

RL with a Search Engine. Following Jin et al. (2025b), we treat the search engine $\mathcal{S}$ as part of the environment rather than as text generated by the policy. The LLM policy π_θ generates trajectories by alternating between reasoning tokens and calls to $\mathcal{S}$. The optimization objective is to maximize expected terminal reward while regularizing the policy with a KL-divergence penalty that limits deviation from a reference policy π_{ref} :

$$\max_{\pi_\theta} \mathbb{E}_{x \sim \mathcal{D}, y \sim \pi_\theta(\cdot|x; \mathcal{S})} [R(x, y)] - \beta D_{\text{KL}} [\pi_\theta(y|x; \mathcal{S}) || \pi_{\text{ref}}(y|x; \mathcal{S})]. \quad (5)$$

Here, $y \sim \pi_\theta(\cdot|x; \mathcal{S})$ denotes a trajectory produced by the joint interaction between policy-generated tokens and environment responses from the search engine $\mathcal{S}$.

Motivation for DSPO. Frameworks such as Search-R1 (Jin et al., 2025b) establish agentic search as an RL problem, but direct application of PPO or GRPO remains difficult in this environment. Open-ended search amplifies the instability of token-level optimization because the reward is assigned to an entire trajectory while the importance ratio is applied to individual tokens (Zheng et al., 2025). This unit mismatch produces high-variance gradient estimates that accumulate across long trajectories and can culminate in policy collapse. At the same time, the sparse binary reward often yields batches containing only successes or only failures; such homogeneous groups produce degenerate advantages and therefore waste samples (Yu et al., 2025; Liu et al., 2025). DSPO is designed to remove both failure modes within a single objective.

3.2 DYNAMIC-FILTER SEQUENCE-LEVEL POLICY OPTIMIZATION

DSPO makes two coordinated modifications to prior value-free policy optimization. First, it performs optimization at the sequence level, aligning the update with the trajectory-level reward. Second, it applies dynamic outcome-based filtering so that each retained rollout group contains non-zero reward variance and therefore yields an informative advantage estimate. Figure 1 summarizes the integrated training process.

3.2.1 SEQUENCE-LEVEL POLICY OPTIMIZATION FOR ENHANCED STABILITY

Following the sequence-level motivation of GSPO (Zheng et al., 2025), DSPO replaces the token-level importance ratio with a sequence-level ratio that matches the reward granularity. For response y_i , the sequence-level importance ratio $s_i(\theta)$ is defined as the geometric mean of its token-level ratios:

$$s_i(\theta) = \left(\frac{\pi_\theta(y_i|x)}{\pi_{\theta_{\text{old}}}(y_i|x)} \right)^{\frac{1}{|y_i|}} = \exp \left(\frac{1}{|y_i|} \sum_{t=1}^{|y_i|} \log \frac{\pi_\theta(y_{i,t}|x, y_{i,<t})}{\pi_{\theta_{\text{old}}}(y_{i,t}|x, y_{i,<t})} \right). \quad (6)$$

This length-normalized ratio reduces variance by keeping $s_i(\theta)$ on a comparable numerical scale across sequences of different lengths, which in turn makes clipping behavior more stable.

Gradient Analysis. The gradient comparison clarifies the stability mechanism. In the unclipped GRPO objective, each token log-probability gradient is multiplied by its own token-specific importance weight $r_{i,t}(\theta)$, allowing local ratio noise to enter the update at every time step. DSPO instead multiplies the sequence log-probability gradient by the single trajectory-level weight $s_i(\theta)$:

$$\nabla_\theta J_{\text{GRPO}} \propto \mathbb{E} \left[\hat{A}_i \cdot \sum_{t=1}^{|y_i|} r_{i,t}(\theta) \nabla_\theta \log \pi_\theta(y_{i,t} | \dots) \right] \quad (7)$$

$$\nabla_\theta J_{\text{DSPO}} \propto \mathbb{E} \left[\hat{A}_i \cdot s_i(\theta) \cdot \sum_{t=1}^{|y_i|} \nabla_\theta \log \pi_\theta(y_{i,t} | \dots) \right] \quad (8)$$

By applying one correction factor to the entire trajectory, DSPO prevents token-level ratio noise from accumulating across long responses. Dynamic filtering complements this sequence-level update by retaining only rollout groups with both successes and failures, thereby ensuring that $\hat{A}_i$ is well defined and non-degenerate in the sparse-reward setting.

3.2.2 DYNAMIC OUTCOME-BASED FILTERING FOR EFFICIENT LEARNING

The binary terminal reward creates a direct challenge for group-relative advantage estimation. When all G responses in a group are correct ($R = 1$) or all are incorrect ($R = 0$), the normalized advantage $\hat{A}_i$ becomes zero or undefined. These homogeneous groups consume rollout computation without providing a useful policy-gradient signal.

DSPO therefore incorporates a dynamic filtering mechanism inspired by DAPO (Yu et al., 2025). During sampling, it retains only trajectory groups containing both successful and unsuccessful outcomes. A group $\{y_i\}_{i=1}^G$ enters the training batch only when its rewards $\{R_i\}_{i=1}^G$ satisfy:

$$0 < \sum_{i=1}^G R_i < G. \quad (9)$$

This condition guarantees non-zero reward variance within every retained training group and therefore produces a meaningful advantage signal. The filter converts sparse terminal feedback into a denser set of informative policy-update comparisons without changing the reward definition itself.

3.3 THE DSPO OBJECTIVE AND TRAINING ALGORITHM

Combining sequence-level ratios with dynamic filtering yields the final DSPO objective. For each valid group sampled from the filtered space $\mathcal{D}_{\text{filtered}}$, DSPO computes $\hat{A}_i$ by group-relative normalization:

$$\hat{A}_i = \frac{R_i - \text{mean}(R)}{\text{std}(R) + \delta}, \quad (10)$$

where δ is a small constant for numerical stability. The policy π_θ is updated by maximizing:

$$J_{\text{DSPO}}(\theta) = \mathbb{E}_{(x, \{y_i\}) \in \mathcal{D}_{\text{filtered}}} \left[\frac{1}{G} \sum_{i=1}^G \min \left(s_i(\theta) \hat{A}_i, \text{clip}(s_i(\theta), 1 - \epsilon, 1 + \epsilon) \hat{A}_i \right) \right], \quad (11)$$

where s_i is the sequence-level importance ratio defined as the geometric mean of the token-level ratios. During likelihood computation, we follow Jin et al. (2025b) and mask all tokens retrieved from the search tool, so the policy is optimized to use external evidence in its reasoning and action decisions rather than to reproduce retrieved text. Algorithm 1 gives the complete training procedure.

Algorithm 1 Dynamic-filter Sequence-level Policy Optimization (DSPO)

```
1: Input: Initial policy  $\pi_{\theta_0}$ , fixed reference policy  $\pi_{\text{ref}}$ , prompt dataset  $\mathcal{D}$ , group size  $G$ , batch size  $B$ , search tool  $\mathcal{R}$ .
2: Initialize policy  $\pi_{\theta} \leftarrow \pi_{\theta_0}$ .
3: for each training step do
4:    $\pi_{\theta_{\text{old}}} \leftarrow \pi_{\theta}$ .
5:   Initialize training buffer  $\mathcal{B} \leftarrow \emptyset$ .
6:   while  $|\mathcal{B}| < B$  do
7:     Sample a prompt  $x \sim \mathcal{D}$ .
8:     Generate a group of  $G$  trajectories  $\{y_i\}_{i=1}^G$  using  $\pi_{\theta_{\text{old}}}$  and the search tool  $\mathcal{R}$ .
9:     Compute terminal rewards  $\{R_i\}_{i=1}^G = \{\text{ContainsAnswer}(y_i, y_{\text{gold}})\}_{i=1}^G$ .
10:    if  $0 < \sum_{i=1}^G R_i < G$  then ▷ Dynamic outcome-based filtering
11:      Add  $(x, \{y_i\}_{i=1}^G, \{R_i\}_{i=1}^G)$  to  $\mathcal{B}$ .
12:    end if
13:  end while
14:  for each  $(x, \{y_i\}, \{R_i\})$  in  $\mathcal{B}$  do
15:    Compute advantages  $\{\hat{A}_i\}_{i=1}^G$  via group normalization of  $\{R_i\}$ .
16:    Compute sequence-level importance ratios  $\{s_i(\theta)\}_{i=1}^G$  using Eq. 6.
17:    Compute the DSPO loss for the group using Eq. 11, applying masks to retrieved tokens.
18:  end for
19:  Update policy parameters  $\theta$  by taking a gradient step on the total loss from  $\mathcal{B}$ .
20: end for
```

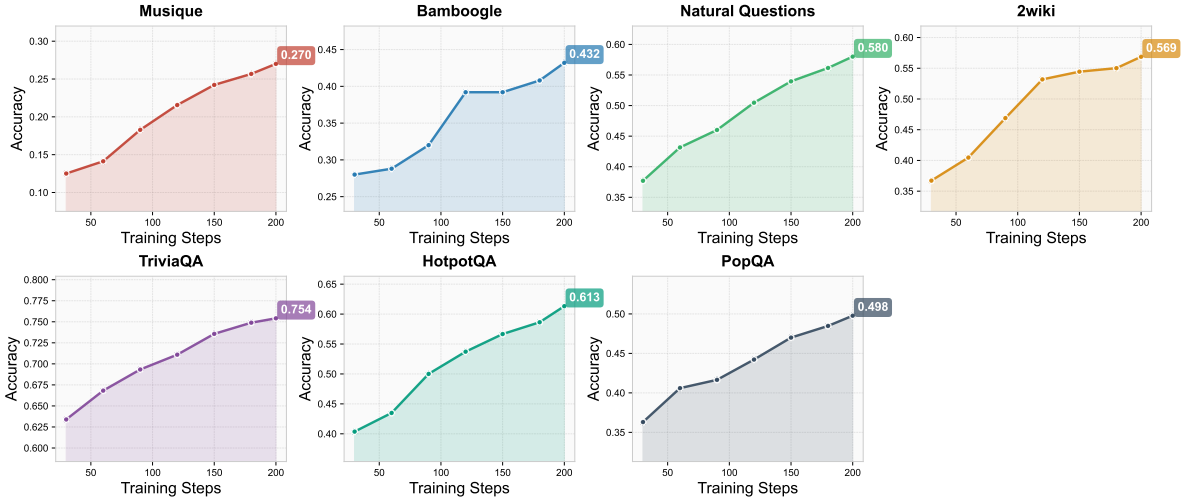


Figure 2: **Validation performance of DSPO across seven benchmarks during training.** Accuracy increases consistently across the evaluated benchmarks, indicating that higher training reward corresponds to improved generalization in the search-and-reasoning policy.

4 EXPERIMENTS

We evaluate DSPO to determine whether its algorithmic design improves both final task performance and the learning dynamics of RL-trained search agents. The experiments address three questions: whether DSPO outperforms strong RL-based search baselines across diverse QA benchmarks; how sequence-level optimization and dynamic filtering each contribute to the full method; and whether the resulting training process is more stable than token-level alternatives. The evidence shows that DSPO’s gains arise from the combination of its two components: sequence-level updates prevent collapse, while dynamic filtering supplies the contrastive reward signal needed for efficient learning.

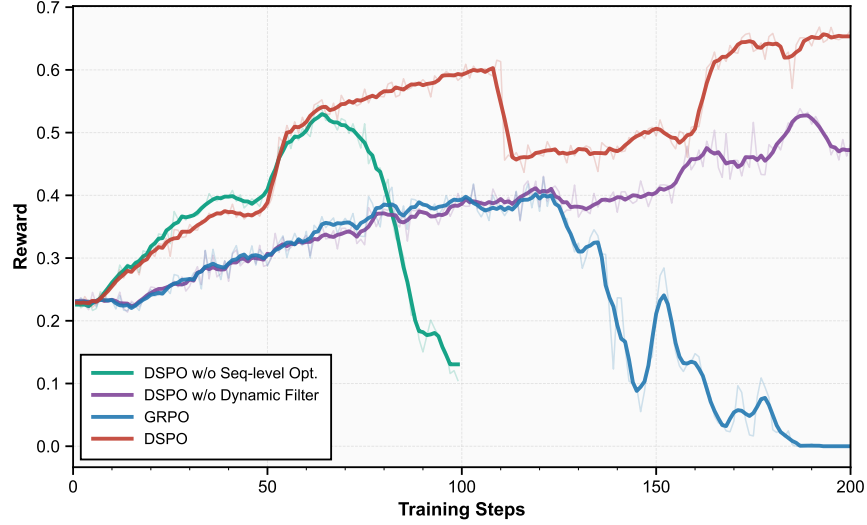


Figure 3: **Training reward dynamics of DSPO and its ablations.** Comparative learning curves show that DSPO (red) improves stably, token-level variants (green, blue) collapse after early gains, and the sequence-level variant without dynamic filtering (purple) remains stable but plateaus at a lower reward.

4.1 EXPERIMENTAL SETUP

Following Search-R1 (Jin et al., 2025b), we use the prompt template in Table 1 to specify the model’s action format during search, including `<think>`, `<tool_call>`, and `<answer>` segments.

Prompt Template. Answer the given question. You must conduct reasoning inside `<think>` and `</think>`. first every time you get new information. After reasoning, if you find you lack some knowledge, you can call a search engine by `<tool_call>` query `</tool_call>` and it will return the top searched results between `<tool_response>` and `</tool_response>`. You can search as many times as your want. If you find no further external knowledge needed, you can directly provide the answer inside `<answer>` and `</answer>`, without detailed illustrations. For example, `<answer>` Beijing `</answer>`. **Question:** ...

Table 1: Prompt template used to structure reasoning, search calls, search responses, and final answers in all experiments.

Benchmarks and Baselines. Our evaluation follows the established Search-R1 protocol (Jin et al., 2025b). We train on a composite dataset consisting of the Natural Questions (NQ) (Kwiatkowski et al., 2019) and HotpotQA (Yang et al., 2018) training splits, then evaluate generalization on seven QA test sets: NQ, TriviaQA (Joshi et al., 2017), PopQA (Mallen et al., 2022), HotpotQA, 2WikiMultiHopQA (Ho et al., 2020), Musique (Trivedi et al., 2022), and Bamboogle (Press et al., 2022). This benchmark suite covers both single-hop factual QA and more complex multi-hop reasoning settings.

The comparison set includes external baselines and diagnostic ablations. External baselines are Qwen2.5-7B and Qwen2.5-14B models trained with PPO and GRPO in the Search-R1 framework (Jin et al., 2025b;a). To isolate DSPO’s mechanisms, we include two ablations: (1) **DSPO w/o dynamic filter**, corresponding to the sequence-level GSPO-style variant (Zheng et al., 2025), and (2) **DSPO w/o sequence-level opt.**, corresponding to the token-level DAPO-style variant (Yu et al., 2025).

For implementation, all 7B training runs start from Qwen2.5-7B-Instruct. The experiments are built on the VeRL framework (Sheng et al., 2025), adapting the provided Search-R1-like code and scripts to our method. Because our experiments use a modified reward function, we retrain the Search-R1 PPO and GRPO baselines (Jin et al., 2025b;a) under the same experimental conditions; these retrained models serve as the primary external comparisons and ensure that the reported differences are attributable to the policy-optimization method rather than to reward or implementation discrepancies.

Implementation and Evaluation. To isolate the contribution of the optimization algorithm, all RL experiments use the same standard BM25 retriever. This controlled retrieval setting prevents improvements from being conflated with retriever strength and attributes performance differences to the learned policy. Across methods, training uses a sparse binary reward based on substring Exact Match (subEM) of the final answer, and subEM is also the primary evaluation metric.

4.2 MAIN RESULTS AND ABLATION STUDY

Table 2 presents the central empirical comparison. Because DSPO’s performance depends on the interaction between sequence-level optimization and dynamic filtering, we report external baselines and ablation variants in the same table. This organization makes the main claim directly testable: DSPO should outperform prior RL-trained search agents while each ablation should reveal the cost of removing one of its stabilizing mechanisms.

Table 2: Comparison of DSPO, Search-R1 baselines, and DSPO ablations on seven QA benchmarks. Baselines include 7B and 14B Search-R1 models trained with GRPO and PPO (Jin et al., 2025b;a); ablations remove either dynamic filtering or sequence-level optimization. Parenthesized values are original Search-R1 EM scores, while non-parenthesized baseline scores are retrained and evaluated under our adjusted reward for consistency. Best results are in **bold**; second-best results are underlined.

Dataset	Search-R1				DSPO & Ablations (Ours, 7B)		
	GRPO (7B)	PPO (7B)	GRPO (14B)	PPO (14B)	w/o dyn. filter	w/o seq-lvl opt.	DSPO
NQ	0.423 (0.429)	(0.393)	<u>0.535</u> (0.482)	(0.424)	0.363	0.470	0.580
TriviaQA	0.658 (0.623)	(0.610)	0.760 (0.667)	(0.660)	0.515	0.695	<u>0.754</u>
PopQA	0.395 (0.427)	(0.397)	<u>0.477</u> (0.434)	(0.442)	0.277	0.430	0.498
HotpotQA	0.401 (0.386)	(0.370)	<u>0.563</u> (0.429)	(0.436)	0.330	0.438	0.613
2WikiMultiHopQA	0.357 (0.414)	(0.346)	0.611 (0.424)	(0.379)	0.285	0.398	<u>0.569</u>
Musique	0.122 (0.162)	(0.146)	<u>0.260</u> (0.191)	(0.210)	0.105	0.133	0.270
Bamboogle	0.280 (0.400)	(0.368)	0.504 (0.492)	(0.480)	0.288	0.280	<u>0.432</u>
Average	0.377 (0.396)	(0.385)	<u>0.530</u> (0.446)	(0.433)	0.313	0.406	0.531

Comparison with Baselines. Table 2 shows that DSPO delivers the strongest overall QA performance among the evaluated 7B methods. The DSPO-trained 7B agent reaches an average subEM score of 0.531, a **34.1% relative improvement** over the same-sized Search-R1 GRPO baseline. It also attains a slightly higher average score than the larger Search-R1 14B models trained with either GRPO or PPO. Because all search experiments use the same BM25 retriever and the principal baselines are retrained under our reward setting, this comparison supports the conclusion that the improvement comes from more effective policy optimization rather than from retrieval strength or model scale alone.

Analysis of Ablations. The ablations in Table 2 identify both components as necessary for DSPO’s final performance. Removing dynamic filtering (‘w/o dynamic filter’) reduces the average score to 0.313, indicating that a sequence-level objective alone cannot learn efficiently when many rollout groups provide no reward contrast. Removing sequence-level optimization (‘w/o sequence-level opt.’) yields an average score of 0.406: this token-level variant benefits from dynamic filtering but remains well below the full DSPO model and, as shown in the next section, is vulnerable to collapse. The pattern establishes the mechanism behind the empirical gains: sequence-level updates provide stability, while dynamic filtering converts sparse terminal rewards into usable group-relative learning signals.

Beyond aggregate metrics, DSPO-trained agents exhibit qualitative search behaviors that are consistent with effective multi-turn policy learning, including recognition of irrelevant retrievals, query reformulation, and follow-up verification (see Appendix A). These behaviors arise from RL training under DSPO rather than from supervised search demonstrations.

4.3 ANALYSIS OF TRAINING DYNAMICS

To evaluate the stability claims directly, we compare the training reward dynamics of DSPO, its ablations, and key baselines in Figure 3. The curves make the role of each algorithmic component visible during optimization rather than only at final evaluation.

DSPO (red) follows a smooth upward trajectory and converges to the highest reward level. The **token-level methods**—DSPO w/o Seq-level Opt. (green) and vanilla GRPO (blue)—briefly improve before collapsing early in training, matching the expected instability from high-variance token-level gradient corrections. **DSPO w/o Dynamic Filter** (purple) remains stable because it retains sequence-level updates, but it plateaus at a substantially lower reward because homogeneous rollout groups provide insufficient learning signal. These dynamics demonstrate that neither component is sufficient in isolation: robust optimization requires both sequence-level stability and dynamic filtering.

We also track validation performance throughout training to verify that reward improvements correspond to benchmark generalization rather than reward overfitting. As shown in Figure 2, DSPO’s validation accuracy on NQ, HotpotQA, and the other evaluated benchmarks increases consistently in parallel with its reward curve. This alignment indicates that the learned policy improves transferable search-and-reasoning behavior under the evaluated QA settings.

4.4 SCALABILITY AND GENERALIZATION ANALYSIS

We further examine whether DSPO’s stability benefits persist beyond the primary 7B QA setting by evaluating parameter-scale transfer and a limited cross-domain generalization setting.

Scalability to Larger Models. We first test whether DSPO’s optimization advantages carry to a larger model by training Qwen2.5-14B-Instruct. As reported in Table 3, the DSPO-trained 14B model reaches an average accuracy of 60.6%, outperforming the GRPO-14B baseline at 53.0% by a relative margin of **14.3%**. The gains appear across all seven benchmarks, indicating that DSPO can exploit increased model capacity while maintaining the stability observed in the 7B setting.

Table 3: Scalability analysis on Qwen2.5-14B-Instruct across the seven QA benchmarks. Gains are relative to the GRPO-14B baseline; best results are in **bold**.

Dataset	Instruct (14B)	GRPO (14B)	DSPO (7B)	DSPO (14B)	Gain
NQ	0.345	0.535	0.580	0.629	+17.6%
HotpotQA	0.407	0.563	0.613	0.665	+18.1%
2WikiMQA	0.332	0.611	0.569	0.699	+14.4%
Bamboogle	0.328	0.504	0.432	0.544	+7.9%
PopQA	0.364	0.477	0.498	0.545	+14.3%
TriviaQA	0.643	0.760	0.754	0.802	+5.5%
Musique	0.151	0.260	0.270	0.361	+38.8%
Average	0.367	0.530	0.531	0.606	+14.3%

Generalization to Mathematical Reasoning. We further test whether the same optimization principles transfer beyond agentic search by applying DSPO to single-turn mathematical reasoning with Qwen2.5 and Qwen3 models. Table 4 reports results on Math500 and Olympiad-Bench. DSPO improves over GRPO for both the 7B and 4B settings, suggesting that the sequence-level and dynamic-filtering principles can benefit reasoning tasks outside retrieval-augmented QA, within the scope of these two evaluated benchmarks.

Table 4: Generalization to mathematical reasoning on Math500 and Olympiad-Bench. DSPO is compared with GRPO under the listed training steps; best results are in **bold**.

Model	Benchmark	Steps	GRPO	DSPO	Gain
Qwen2.5-Math-7B	Math500	200	0.772	0.798	+2.6%
Qwen3-4B	Olympiad-Bench	100	0.728	0.755	+2.7%

5 CONCLUSION

This work addresses the instability and sample inefficiency that limit RL training for autonomous LLM search agents. We introduced Dynamic-filter Sequence-level Policy Optimization (DSPO), which stabilizes training by aligning policy updates with sequence-level rewards and by filtering rollout groups so that sparse terminal rewards yield informative advantages. Across challenging QA benchmarks, DSPO delivers stronger final performance than the evaluated PPO and GRPO search baselines while exhibiting substantially more stable training dynamics.

By learning from environmental feedback alone, DSPO provides a practical route to training capable search agents without supervised expert trajectories. The results also clarify a general design principle for RL with LLM agents: the optimization unit should match the reward unit, and retained samples should contain enough outcome variation to support advantage estimation. Future work can build on this foundation by pairing DSPO with stronger retrievers, extending it to multi-tool settings, and validating its principles more broadly in domains such as mathematics and code generation.

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A APPENDIX

Table A.1 provides representative trajectories that illustrate the search policies learned by DSPO. The examples show three behaviors that support the quantitative results: reformulating queries after irrelevant retrievals, verifying uncertain information through follow-up searches, and extracting final answers from retrieved evidence.

Table A.1: Representative trajectories illustrating learned search and reasoning behavior. The examples show (a) recognition of irrelevant retrievals, (b) query reformulation after an initial failure, and (c) verification before final-answer generation.

Example 1: Presidential Succession (<i>Query Refinement after Failure</i>)	
Question	Who is next in line if something happens to the president?
Gold	vice president
Turn 1	<code>search("who is next in line if something happens to the president")</code> → Retrieved: Documents about novels "Something Happened" and "Mosquitoes Buzz" Agent: "The queries were not directly about the line of succession... The relevant information I need is not present in the search results. Let's try searching again with a more specific query."
Turn 2	<code>search("line of succession for the president")</code> → Retrieved: Philippine and US presidential succession information Agent: "Based on the search results... the next person in line would be the Vice President."
Answer	✓ The Vice President
Example 2: University Identification (<i>Information Verification</i>)	
Question	Which American University is situated in Ithaca, New York?
Gold	Cornell University
Turn 1	<code>search(["American University in Ithaca", "University in Ithaca NY"])</code> → Retrieved: "Ithaca is home to Cornell University, an Ivy League school..." Agent: "Ithaca, New York is home to Cornell University, which is an Ivy League school. However... it is highly likely that the university in question is Cornell. Let me confirm this."
Turn 2	<code>search(["Cornell University location", "Cornell University New York"])</code> → Retrieved: Cornell history confirming establishment in Ithaca Agent: "Cornell University is indeed located in Ithaca, New York."
Answer	✓ Cornell University
Example 3: Biographical Information (<i>Direct Extraction</i>)	
Question	What is John Andrew Martin's occupation?
Gold	lawyer, attorney, politician
Turn 1	<code>search(["John Andrew Martin occupation", "John Andrew Martin job"])</code> → Retrieved: "American journalist, attorney, soldier, and politician..." Agent: "John Andrew Martin was an American <u>journalist, attorney, soldier, and politician</u>. He represented Colorado in the U.S. House of Representatives."
Answer	✓ Journalist, attorney, soldier, and politician